Sec 11.3 Partial Derivatives

What is a derivative? It is a rate of change of one quantity with respect to another. But what if our function depends on more than one variable?

Temperatures Delivered by Evaporative Coolers

% Relative Humidity

Air Temperature	2	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80
75	54	55	57	58	59	61	62	63	64	65	66	67	68	69	70	71	72
80	57	58	60	62	63	64	66	67	68	69	71	72	73	74	76	76	77
85	61	62	63	65	67	68	70	71	72	73	74	75	76	77	79	81	
90	64	65	67	69	70	72	74	76	77	78	79	81	82	83	84	86	
95	67	68	70	72	74	76	78	79	81	82	84	85	87				
100	69	71	73	76	78	80	82	83	85	87	88						
105	72	74	77	79	81	84	86	88	89								
110	75	77	80	83	85	87	90	92									
115	78	80	83	86	89	91	94										
120	81	83	86	90	93	95											
125	83	86	90	93	96												

Optimum conditions for Evaporative Coolers

Source: Ed Phillips, Arizona Almanac

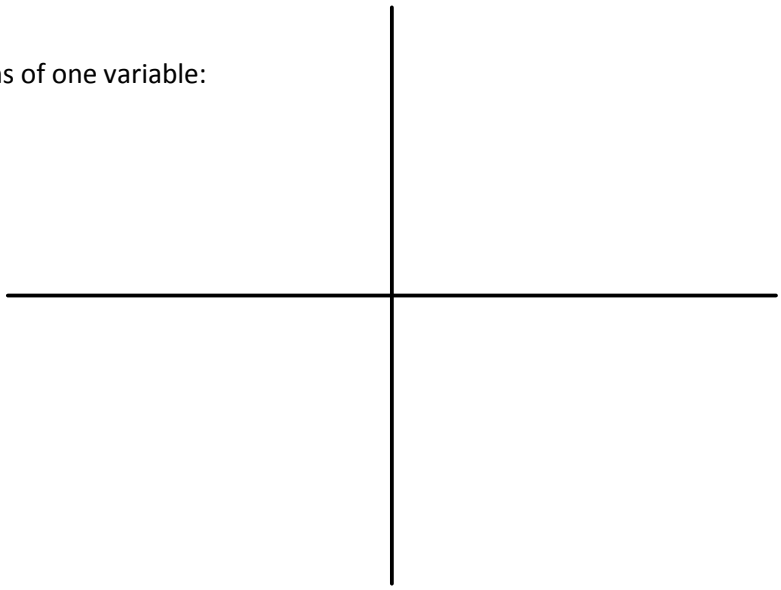
A partial derivative is the rate of change of a multivariable function in which all but one of the variables remain fixed. It is the rate of change of a function f with respect to one of its variables only.

Sec 11.3 Partial Derivatives

How do we find partial derivatives? How are they defined?

You may recall the definition of a derivative for functions of one variable:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$



For a partial derivatives

4 If f is a function of two variables, its **partial derivatives** are the functions f_x and f_y defined by

$$f_x(x, y) = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h}$$

$$f_y(x, y) = \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h}$$

That is we treat one variable as a constant.

For Functions of more than two variables the definition is the same.

$$\frac{\partial u}{\partial x_i} = \lim_{h \to 0} \frac{f(x_1, \ldots, x_{i-1}, x_i + h, x_{i+1}, \ldots, x_n) - f(x_1, \ldots, x_i, \ldots, x_n)}{h}$$

Notations for Partial Derivatives If $z = f(x, y)$, we write

$$f_x(x, y) = f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x, y) = \frac{\partial z}{\partial x} = f_1 = D_1 f = D_x f$$

$$f_y(x, y) = f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} f(x, y) = \frac{\partial z}{\partial y} = f_2 = D_2 f = D_y f$$

Sec 11.3 Partial Derivatives

Rule for Finding Partial Derivatives of $z = f(x, y)$

1. To find f_x, regard y as a constant and differentiate $f(x, y)$ with respect to x.
2. To find f_y, regard x as a constant and differentiate $f(x, y)$ with respect to y.

HW example

15–38 Find the first partial derivatives of the function.

15. $f(x, y) = y^5 - 3xy$

29. $f(x, y, z) = xz - 5x^2y^3z^4$

31. $w = \ln(x + 2y + 3z)$

Sec 11.3 Partial Derivatives

Geometrically, what are these partial derivatives? Consider the function $z = 4 - x^2 - 2y^2$

Let's find $f_x \; and \; f_y$ at the point (1,1)

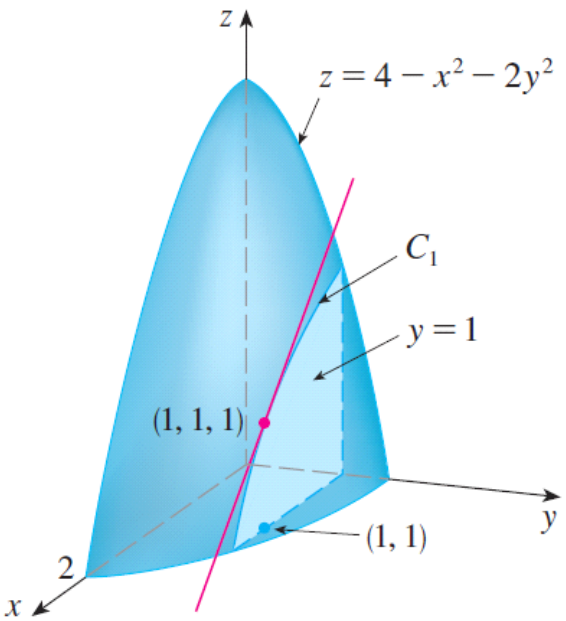


FIGURE 2

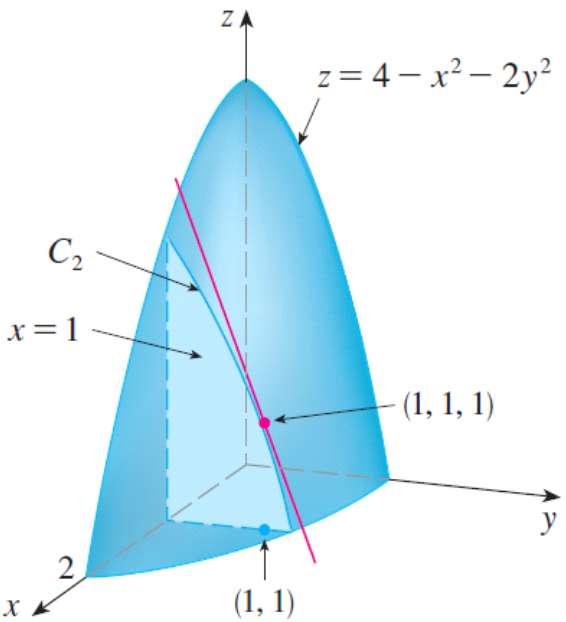


FIGURE 3

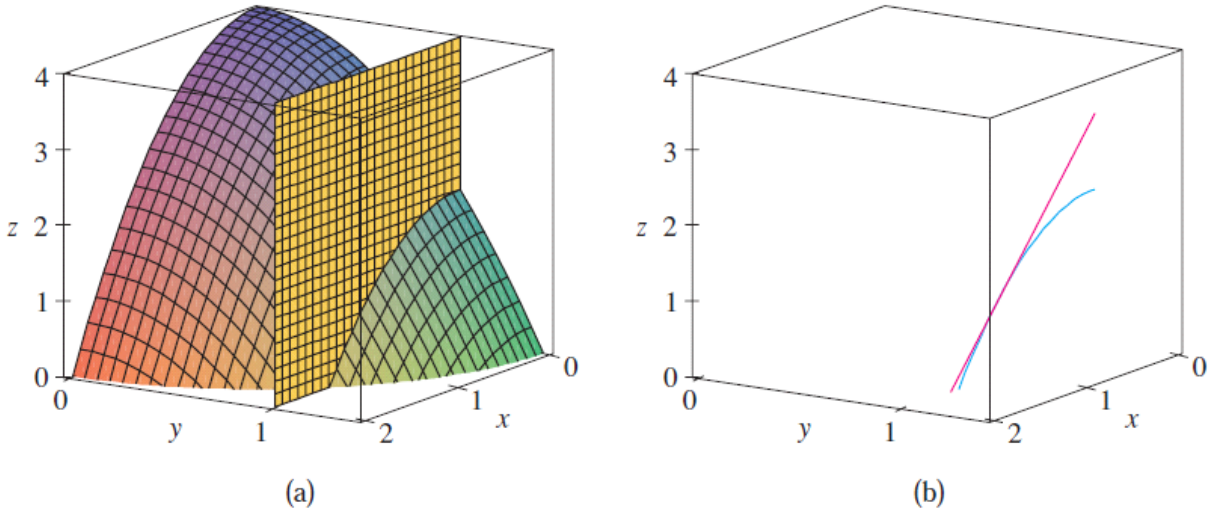


FIGURE 4

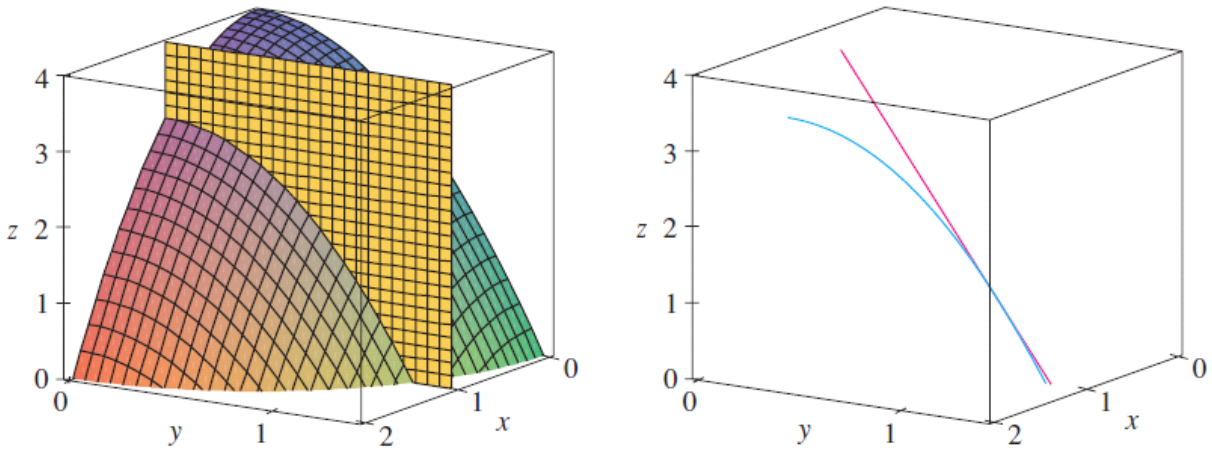


FIGURE 5

Sec 11.3 Partial Derivatives

Higher partial derivatives. If we can take a partial once can we do it again?

The second partial with respect to x ----> $$(f_x)_x = f_{xx} = f_{11} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 z}{\partial x^2}$$

The second partial with respect to x then to y---> $$(f_x)_y = f_{xy} = f_{12} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 z}{\partial y\,\partial x}$$

The second partial with respect to y then to x---> $$(f_y)_x = f_{yx} = f_{21} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 z}{\partial x\,\partial y}$$

The second partial with respect to y----> $$(f_y)_y = f_{yy} = f_{22} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 z}{\partial y^2}$$

HW Example

51–56 Find all the second partial derivatives.

51. $f(x, y) = x^3y^5 + 2x^4y$

Sec 11.3 Partial Derivatives

Clairaut's Theorem Suppose f is defined on a disk D that contains the point (a, b). If the functions f_{xy} and f_{yx} are both continuous on D, then

$$f_{xy}(a, b) = f_{yx}(a, b)$$

Hw example

57–58 Verify that the conclusion of Clairaut's Theorem holds, that is, $u_{xy} = u_{yx}$.

57. $u = xe^{xy}$

Sec 11.3 Partial Derivatives

Even higher derivatives

59–64 Find the indicated partial derivative(s).

59. $f(x, y) = 3xy^4 + x^3y^2; \quad f_{xxy}, \quad f_{yyy}$

61. $f(x, y, z) = \cos(4x + 3y + 2z); \quad f_{xyz}, \quad f_{yzz}$

Partial Differential Equations

Partial derivatives occur in *partial differential equations* that express certain physical laws. For instance, the partial differential equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

is called **Laplace's equation** after Pierre Laplace (1749–1827). Solutions of this equation are called **harmonic functions**; they play a role in problems of heat conduction, fluid flow, and electric potential.

HW example:

73. If f and g are twice differentiable functions of a single variable, show that the function

$$u(x, t) = f(x + at) + g(x - at)$$

The **wave equation**

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$$